Ola Ride Insights

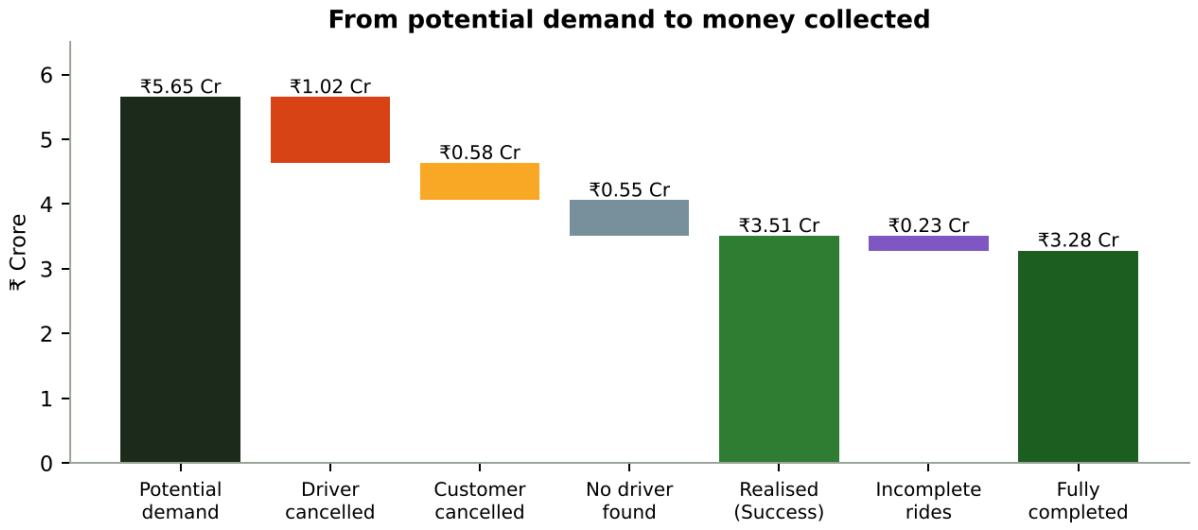
Business Intelligence report: 103,024 bookings, 1–31 July 2024, Bengaluru

Bookings	Success rate	Revenue realised	Revenue leaked (% of demand)	Customers
103,024	**62.1%**	**₹3.51 Cr**	**₹2.15 Cr · 38%**	**94,544**

Executive summary

- **Only 62.1% of bookings turn into a ride.** 39,057 bookings failed, leaking ₹2.15 Cr, which is 38% of the money customers tried to spend.
- **Drivers are the biggest leak.** Driver cancellations alone cost ₹1.02 Cr. Another 56% of the 'customer' cancellations were triggered by the driver (not moving to pickup, or asking the customer to cancel).
- **'Success' is not the same as 'completed'.** 3,926 rides inside the Success status ended incomplete (₹22.50 L), so fully completed revenue is ₹3.28 Cr.
- **The failure rate is system-wide, not a hotspot.** Vehicle type, hour, weekday, pickup area and route show no meaningful difference in success rate (all chi-square tests are noise).
- **Retention is weak.** 92% of customers booked once, and 34,557 customers (37%) never completed a single ride. 1,725 of them tried at least twice.
- **Cash still leads:** 55% of completed rides, UPI 40%, cards 5%.

Revenue waterfall



Read this first. The dataset behaves like **synthetic data**: fares, ratings and wait times are statistically unrelated to distance, vehicle or each other. The methods and dashboards in this report are valid, but the numbers should not be quoted as real facts about Ola.

1. Data understanding & preprocessing

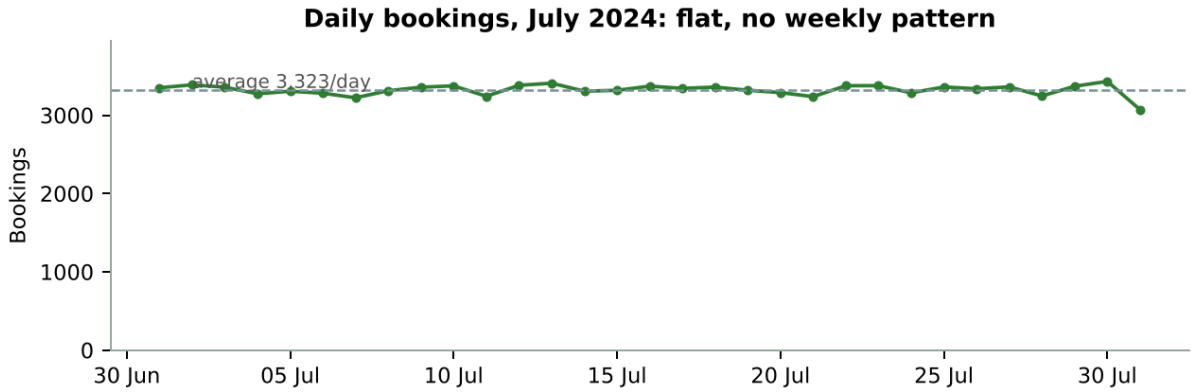
Source: **OLA_DataSet.xlsx**, sheet July (103,024 rows × 20 columns). Sheets Sheet1/Sheet2 only hold a scratch pivot and vehicle icon URLs. The script **clean_data.py** performs every step below and writes the log to **cleaning_log.csv**. **No rows were dropped and nothing was imputed.**

Step	What was found	What was done
Duplicates	Booking_ID is unique; no duplicate rows	Nothing to remove
Date / Time	Date held date + clock time and matched the Time column on 100% of rows	Split into Date, Time, Booking_Datetime
Naming	'Vehicle Images' (space); Canceled_Rides_by_* actually hold reasons	Renamed to Vehicle_Image_URL, Customer_Cancel_Reason, Driver_Cancel_Reason, Driver_Rating
Categories	'eBike' vs 'E-Bike' in the slides	Standardised to 'E-Bike'
Blank cells	39,057 blank payment, rating and TAT cells, all on non-Success rides	Structural, not missing: a ride that never happened has no payment or rating. Text blanks became 'Not Applicable'; numeric blanks stayed empty (imputing would invent data)
Consistency	Non-Success rides always have 0 km; Success rides always have payment, rating and TAT	Validated with assertions in the script
Outliers	Booking_Value ₹100–₹2,999, right-skewed (median ₹386, mean ₹549); ratings 3.0–5.0	Kept. Values are plausible fares, not errors; high fares are flagged separately
Anomalies	2,121 rides with same pickup and drop; 640 above P99 fare per km; 442 short rides (≤3 km) priced ≥ ₹1,000	Flagged with Flag_* columns, not deleted

Derived features (22 new columns)

Hour, Day_Name, Is_Weekend, Week, Time_Band · Is_Success, Is_Incomplete, Is_Fully_Completed, Cancelled_By, Failure_Type · Revenue_Realised, Revenue_Lost · Route, Rating_Gap, Fare_per_km, Distance_Band, Fare_Band · Flag_Same_Pickup_Drop, Flag_Extreme_Fare_per_km, Flag_Short_Ride_High_Fare, Anomaly_Score.

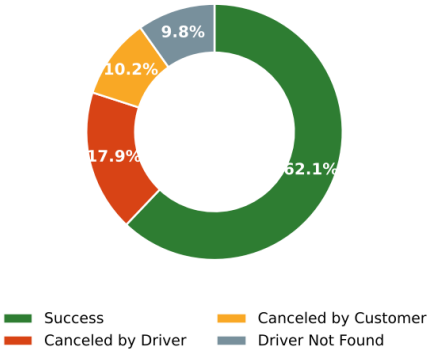
Calendar trap avoided. Mon/Tue/Wed appear five times in July 2024 and every other weekday four times, so raw weekday totals look 25% higher on Mon–Wed. Per calendar day, demand is flat at about 3,300 bookings (range 3,072 to 3,432).



2. Where the money leaks

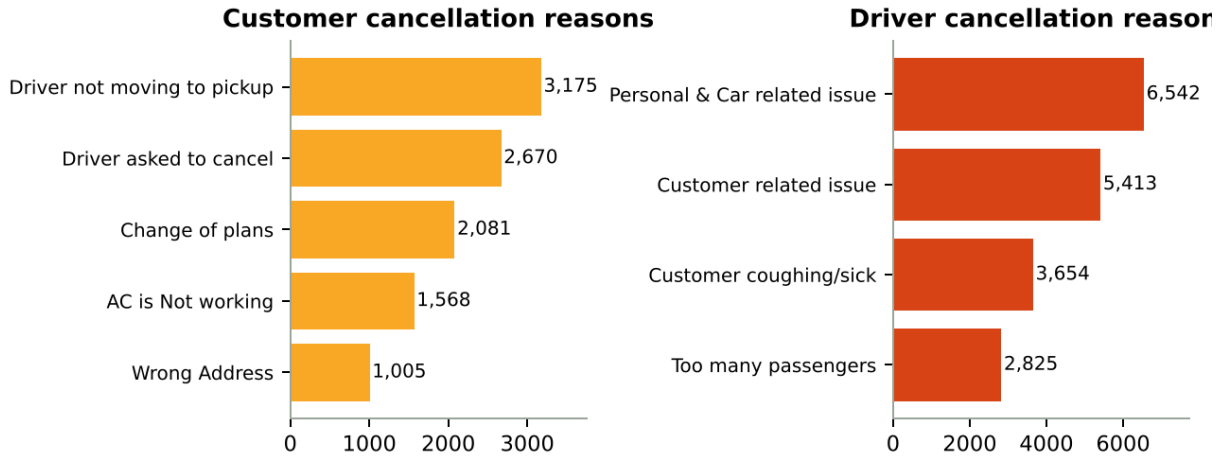
A failed booking carries the Booking_Value the customer was quoted, which is treated here as revenue that would have been earned. Of ₹5.65 Cr in potential demand, ₹3.51 Cr was realised.

Booking outcomes (103,024)



Outcome	Bookings	Value
Success	63,967	₹3.51 Cr
Canceled by Driver	18,434	₹1.02 Cr
Canceled by Customer	10,499	₹0.58 Cr
Driver Not Found	10,124	₹0.55 Cr
Incomplete (inside Success)	3,926	₹0.23 Cr

Why cancellations happen

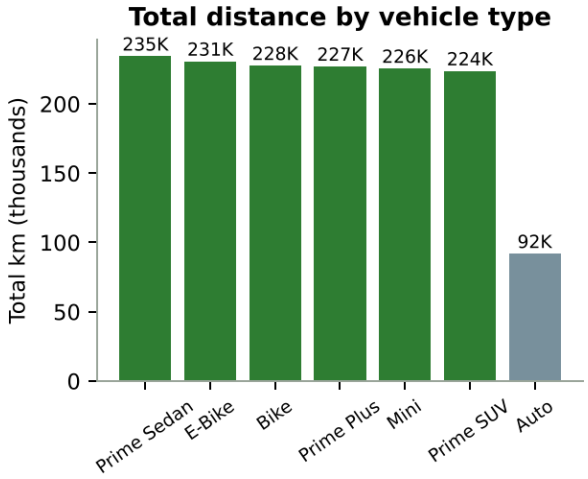


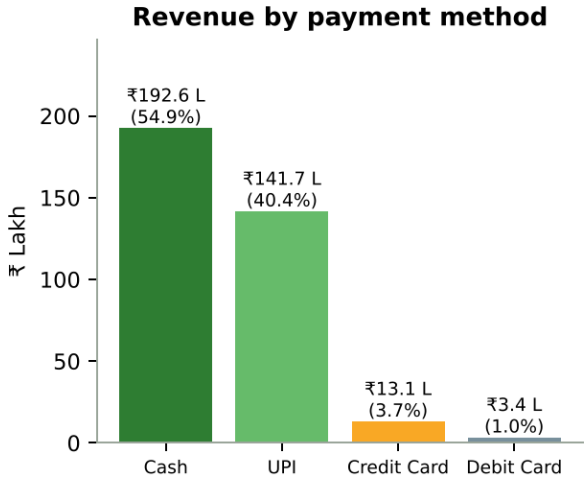
- **Driver cancellations (18,434):** 'Personal & car related issue' is the top reason (6,542, 35%). Rider-related reasons (customer issue, coughing/sick, too many passengers) make up the other 65%.
- **Customer cancellations (10,499):** the top two reasons, 'driver not moving towards pickup' and 'driver asked to cancel', account for 5,845 (56%). These are driver-caused even though they are labelled customer cancellations.
- **No driver found (10,124, ₹0.55 Cr):** a supply gap. Success rate does not vary by area, so the shortage is city-wide.
- **Incomplete rides (3,926):** Customer Demand 1,601, Vehicle Breakdown 1,591, Other 734.

3. Vehicles, payments and customers

Vehicle scorecard

Vehicle	Total booking value	Success booking value	Avg km (completed)	Total km	Success rate
Prime Sedan	₹82.99 L	₹52.24 L	25.0	234,535	63.0%
E-Bike	₹81.82 L	₹50.55 L	25.1	230,842	62.0%
Auto	₹80.92 L	₹50.52 L	10.0	92,043	62.1%
Prime Plus	₹80.51 L	₹50.15 L	25.0	227,186	61.7%
Bike	₹79.87 L	₹49.72 L	24.9	227,746	62.3%
Mini	₹79.91 L	₹48.86 L	25.0	225,703	62.1%
Prime SUV	₹79.33 L	₹48.77 L	24.9	223,848	61.4%



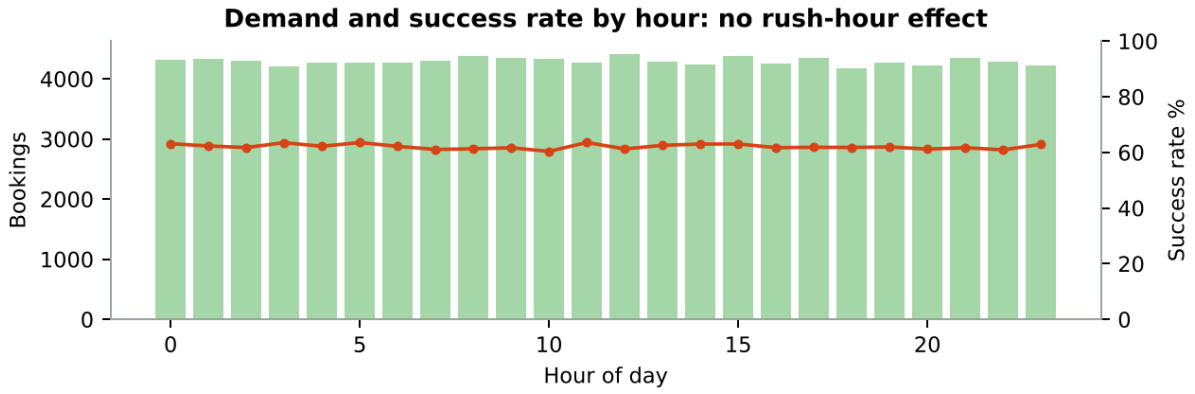


- All six non-Auto vehicles are almost identical: about 25 km average, success rate 61–63%. **Autos are capped at 19 km (average 10 km)**, so they carry only about 40% of the distance of the other types despite earning similar revenue (₹50.52 L). Bike and E-Bike rides of up to 49 km are unusually long and worth checking against business rules.
- **Payments:** Cash ₹192.60 L (54.8% of rides), UPI ₹141.68 L (40.5%), credit card 3.8%, debit card 1.0%. Average fare is the same across methods (about ₹540–550).

Customers

- 94,544 unique customers; 86,590 (91.6%) booked only once and 7,954 (8.4%) booked 2+ times. The heaviest user booked 5 rides.
- **34,557 customers (36.6%) never completed a ride.** Of these, **1,725** tried at least twice and still never got a ride: the highest-value win-back audience.
- Ratings are uniform between 3.0 and 5.0 (mean 4.0 for both drivers and customers) with no 1–2 star rides and no correlation between driver and customer ratings ($r \approx 0$). Ratings in this extract cannot identify problem drivers.

4. Demand patterns, pricing and anomalies



Signal vs noise: success rate tested against each factor (chi-square, all rows)

Factor	p-value	Effect size (Cramér's V)	Verdict
Vehicle_Type	0.119	0.0099	Noise
Hour	0.070	0.0181	Noise
Time_Band	0.492	0.0065	Noise
Day_Name	0.748	0.0058	Noise
Is_Weekend	0.561	0.0018	Noise
Pickup_Location	0.730	0.0203	Noise
Drop_Location	0.456	0.0219	Noise

- **Peak-hour optimisation has no data support here.** Demand and success rate are flat across all 24 hours, so driver allocation by hour cannot be justified from this extract.
- **Pricing:** fare is statistically independent of distance (r = 0.002) and vehicle type. Median fare per km is ₹144 for 1–5 km rides but only ₹9.6 for 31–50 km rides. Surge-pricing effectiveness cannot be measured; check the fare source system.
- **Fraud / anomaly watch:** 2,833 rides carry at least one flag (2,121 same pickup and drop, 640 fare per km above the 99th percentile of ₹641/km, 442 rides of ≤3 km priced ≥ ₹1,000). Total booking value at stake: ₹0.23 Cr. These are review candidates, not confirmed fraud.

5. Recommendations

Impact figures are estimates: a 10% reduction in each bucket, assuming recovered bookings convert at that bucket's average value.

#	Action	Evidence	Est. gain per 10% fix
1	Cut driver cancellations: pre-trip vehicle checks, acceptance incentives, penalties for repeat cancellers, capture passenger count at booking	18,434 cancellations, ₹1.02 Cr; top reason is personal/car issues	₹10.18 L
2	Auto-reassign when the assigned driver is not moving toward pickup; track driver ETA movement	56% of customer cancellations are driver-triggered	₹5.77 L
3	Add supply city-wide (incentives, surge-time bonuses) rather than targeting single areas	10,124 no-driver bookings; no area effect	₹5.50 L
4	Reduce breakdowns with vehicle health checks; confirm destination before the trip	3,926 incomplete rides; 40% breakdown, 41% customer demand	₹2.25 L
5	Win-back campaign (vouchers + guaranteed priority) for customers whose bookings all failed	1,725 customers with 2+ failed attempts; 34,557 never completed a ride	Retention
6	Push UPI with small cashback to move cash share down	Cash 55% of completed rides	Cost / cash handling
7	Audit fare engine and review flagged rides	Fare unrelated to distance; 2,833 flagged rides	Risk reduction

Limitations

- One month of data, one city; no driver IDs, surge multipliers, ETA or timestamps for cancellation, so supply and pricing questions can only be partly answered.
- Booking_Value of failed bookings is a quoted value, not money collected.
- TAT columns (V_TAT, C_TAT) have no unit in the source, so they are not interpreted here.
- Almost no variable relates to another, which suggests simulated data. Use the findings to demonstrate method and storytelling.

6. SQL query results

Full queries (plus 5 bonus BI queries) are in **ola_queries.sql**. Every result was cross-checked against pandas.

#	Question	Result
1	All successful bookings	63,967 rows
2	Average ride distance per vehicle type	E-Bike 25.15 · Prime Plus 25.03 · Prime Sedan 25.01 · Mini 24.98 · Bike 24.93 · Prime SUV 24.88 · Auto 10.04 km
3	Rides cancelled by customers	10,499
4	Top 5 customers by number of rides	CID954071 (5 rides), then 4 customers with 4 rides each (ties broken by total booking value)
5	Driver cancellations: personal & car-related	6,542
6	Prime Sedan driver rating, max and min	5.0 and 3.0
7	Rides paid by UPI	25,881 rows
8	Average customer rating per vehicle type	Prime Plus 4.009 · Prime Sedan 4.002 · Auto 3.999 · Prime SUV 3.999 · Mini 3.998 · Bike 3.993 · E-Bike 3.988
9	Total booking value of successful rides	₹35,080,467 (₹3.51 Cr); fully completed only: ₹32,829,985
10	Incomplete rides with reason	3,926 rows (Customer Demand 1,601 · Vehicle Breakdown 1,591 · Other Issue 734)

Sample query (Q2)

```
SELECT Vehicle_Type,
       ROUND(AVG(Ride_Distance), 2) AS avg_distance_km
FROM rides
WHERE Booking_Status = 'Success'
GROUP BY Vehicle_Type
ORDER BY avg_distance_km DESC;
```

7. Power BI dashboard blueprint

Power BI Desktop is a Windows app, so the .pbix file has to be built on your PC. This blueprint matches your slide layout (5 pages with a navigation strip) and the 10 required visuals.

Setup

- Open Power BI Desktop → **Get data → Text/CSV** → select **ola_clean_for_powerbi.csv** → Load. Rename the table to **rides**.
- In Power Query check types: Date = Date, Booking_Datetime = Date/Time, Time = Text, Hour/Ride_Distance/Booking_Value = Whole number, ratings = Decimal, flags = True/False.
- Optional vehicle icons: import the 7 icon URLs from Sheet2 of your Excel as a second table, relate it to rides on Vehicle_Type, and set the URL column's data category to **Image URL**. Use it in the vehicle table on the Vehicle Type page.
- Insert → **Buttons → Page navigator** on every page for the Overall / Vehicle Type / Revenue / Cancellation / Ratings strip, as in your slides. Add slicers for Date, Vehicle_Type and Time_Band and sync them across pages (View → Sync slicers).

DAX measures (Home → New measure)

```
Total Bookings        = COUNTROWS(rides)
Success Bookings      = CALCULATE(COUNTROWS(rides), rides[Booking_Status] = "Success")
Success Rate          = DIVIDE([Success Bookings], [Total Bookings])
Total Booking Value   = SUM(rides[Booking_Value])
Success Booking Value = CALCULATE(SUM(rides[Booking_Value]), rides[Booking_Status] = "Success")
Revenue Leaked        = [Total Booking Value] - [Success Booking Value]
Avg Distance (km)     = CALCULATE(AVERAGE(rides[Ride_Distance]), rides[Booking_Status] = "Success")
Total Distance (km)   = SUM(rides[Ride_Distance])
Avg Customer Rating   = AVERAGE(rides[Customer_Rating])
Avg Driver Rating     = AVERAGE(rides[Driver_Rating])
Cancelled by Customer = CALCULATE(COUNTROWS(rides), rides[Booking_Status] = "Canceled by Customer")
Cancelled by Driver   = CALCULATE(COUNTROWS(rides), rides[Booking_Status] = "Canceled by Driver")
```

Design tips

- Use one colour per outcome everywhere: Success green, Driver cancel red, Customer cancel amber, No driver found grey, Incomplete violet.
- Put the 4 KPI cards (Bookings, Success Rate, Revenue Leaked, Avg Fare) at the top of the Overall page so the story starts with the leak.
- Always filter 'Not Applicable' out of reason and payment visuals, otherwise a grey bar of 'nothing' dominates the chart.
- Add tooltips with Success Rate and Revenue Leaked to the vehicle and area visuals, and sync the Date slicer on all pages.

Publish and embed: Home → Publish → Power BI Service → File → Embed report → Publish to web. Paste the link in the **Power BI** tab of the Streamlit app. Theme tip: View → Themes → Customize, background dark green #1B2A1B, accent #C6F432.

Pages and visuals

Page	Visual (brief question)	How to build it
Overall	1. Ride Volume Over Time	Line/column chart: X = Date, Y = Total Bookings
Overall	2. Booking Status Breakdown	Donut: Legend = Booking_Status, Values = Total Bookings. Add 4 KPI cards: Total Bookings, Success Rate, Success Booking Value, Revenue Leaked
Vehicle Type	3. Top 5 Vehicle Types by Ride Distance	Bar chart: Y = Vehicle_Type, X = Total Distance; Filters → Top N = 5 by Total Distance. Add a Table with the 7 vehicles: Total Booking Value, Success Booking Value, Avg Distance, Total Distance (as on your slide 2)
Revenue	6. Revenue by Payment Method	Donut or bar: Legend = Payment_Method, Values = Success Booking Value; filter out 'Not Applicable'
Revenue	7. Top 5 Customers by Total Booking Value	Bar: Y = Customer_ID, X = Success Booking Value; Top N = 5
Revenue	8. Ride Distance Distribution Per Day	Column chart: X = Date, Y = Avg Distance (or box plot via the Box and Whisker custom visual)
Cancellation	5. Canceled Rides Reasons	Two bar charts: Customer_Cancel_Reason and Driver_Cancel_Reason by Total Bookings; filter out 'Not Applicable'
Ratings	4. Avg Customer Rating by Vehicle Type	Clustered column: X = Vehicle_Type, Y = Avg Customer Rating (set axis min 3.5)
Ratings	9. Driver Ratings Distribution	Column chart: X = Driver_Rating, Y = Count of Booking_ID
Ratings	10. Customer vs Driver Ratings	Scatter: X = Avg Customer Rating, Y = Avg Driver Rating, Details = Vehicle_Type. Or two histograms side by side